Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*			higher yield would be formed using a lower temperature (1) however lower temperatures result in a lower reaction rate (1) use of catalyst increases rate compensating for use of a moderately low temperature (1)	3			3		
	(b)			$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$ award (1) for product award (1) for balancing **only** if all formulae are correct		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(c)*			**Indicative content** place sulfuric acid in burette measure 25 cm^3 of ammonium hydroxide (into conical flask) add few drops of indicator e.g. phenolphthalein add acid steadily until end-point approaches and drop-wise near end-point record volume of acid needed to just change indicator colour solution is neutral - but contaminated with indicator repeat without indicator - measure 25 cm^3 of ammonium hydroxide (to clean flask) and add exactly the volume of sulfuric acid required to neutralise the alkali solution is neutral - only ammonium sulfate and water present boil off some of the water and leave to cool forming crystals / leave solution to evaporate slowly to form crystals overnight dry crystals (if necessary) sequenced labelled diagrams and appropriate equations should be credited marks limited to lower band if insoluble oxide/carbonate method given **5-6 marks** Full description and explanation of each stage; good attempt at equations *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** Description and partial explanation of at least two stages *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Basic description of neutralisation and crystallisation *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*	6			6		6
				Question 7 total	**9**	**2**	**0**	**11**	**0**	**6**